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About Me

Driven by innovation, electrical engineer with expertise in robotics. Two years+ experience working with Python and C++ for drone-based inspection and aerial manipulation.

Contact

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-  digofbastos@gmail.com
-  www.linkedin.com/in/rodrigo-freire-bastos
-  Salvador, BA

Skills

- Linux (Ubuntu)
- Python
- C++
- PX4-Autopilot
- Robotics
- Electronics

Language

- English (Advanced)
- German (Basics)

RODRIGO BASTOS

Electrical Engineering

Work Experience

Robotics Engineer Intern

2024-2025

SENAI CIMATEC

Simulated UAVs and soft robots in Gazebo with ROS2, integrated PX4-Autopilot for autonomous missions, and developed robotic manipulation systems. Worked with embedded systems, sensor fusion, and real-time data. Applied C++, Python, and Linux tools for robotic software development.

Robotics Junior Researcher

2023 - 2024

SENAI CIMATEC

Junior robotics researcher at SENAI CIMATEC. Conducted applied research in the field of aerial robotics, sensor fusion and perception for autonomous mapping and navigation. Results of this work were published and presented at SIINTEC X.

Scientific Research Volunteer

2021-2023

SENAI CIMATEC

Fellow researcher in a CNPq project, I developed an IoT solution with thermal camera computer vision for automatic leak detection in onshore oil pipelines. This system aims to enhance real-time monitoring, providing early and reliable detection of leaks in remote areas.

Personal Project

Solarfly Drone

Innovative solution to revolutionize inspection and cleaning at solar plants through the use of autonomous drones

Education

SENAI CIMATEC

2021-2025

Bachelor of Electrical Engineering

Bibliographical Production

Map reconstruction and Localization: A comparison of LiDAR Odometry Algorithms for UAV Navigation

LIO; Drone; SLAM; Mapping; Localization

IoT Leak Detection System for Onshore Oil Pipeline Based on Thermography

leak detection; onshore oil fields; Internet of Things